

The 20-Round Barrier for Table-Based Preimage Attacks on SHA-256

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Abstract

This is a companion to [1], which gives a practical oracle-free preimage attack on 19-round SHA-256 built on a precomputed $\sigma_0(u)-u$ differential table. Here we ask why the same construction does not reach 20 rounds, and characterise the barrier quantitatively rather than merely reporting it.

The $R=19$ attack is shown not to schedule its constraints acyclically: it breaks a cycle by freezing W_9 provisionally, iterating, and paying one self-consistency check. Promoting a_{11} to an unknown absorbed by the fourth constraint gives $R=20$ the same one-word surplus, and the true $R=20$ solution is verified to be a fixed point of all four absorber maps—so the architecture transplants in form. It does not transplant in efficacy: measured with a single harness, the $R=19$ two-map iteration converges at 5.34×10^{-5} per a_0 (1,790 of 3.36×10^7) while the $R=20$ four-map iteration yields 0 of 2.00×10^9 ($< 1.50 \times 10^{-9}$ at 95%), below the 2^{-29} break-even. The cause is dimensional: the iterated state grows from 32 to 64 bits.

We then give the structural reason. Each free state word a_k enters exactly two recovered message words linearly, W_k with coefficient $+1$ and W_{k+8} with coefficient -1 , so constraint C_j absorbs a_{j+1} through the one global $\sigma_0(u)-u$ table for *every* j , $C3 \rightarrow a_4$ included. Among the absorbed words, dependence is full-avalanche only in the forward direction. At $R=19$ every feedback edge passes through Maj/Ch alone (at most 1.9 bits of a constraint change per flipped bit); at $R=20$ exactly one feedback edge is heavy, $a_4 \rightarrow C0$ through W_9 (12.2 bits), and the heavy dependency graph acquires its first cycle. The $R=19$ iteration works because it cuts its cycle at a mild edge; a heavy cycle has no mild cut, and any 32-bit cut of it is a random map. The obstruction is $\Sigma_0(a_4)$ subtracted directly inside W_9 , which no context choice removes; it is the reach of the round function, not of the frame: the window is all-mild exactly when $R \leq 19$, and $R=21$ has three heavy feedback edges. We give the corrected inventory of which absorptions admit a global table, show the assignment is forced up to a choice between two frames, both of which close the heavy cycle, and record the resulting 2^{64} cost for this attack family. We record a second global table (the a_0 absorber, $\Gamma(a_0) = a_0 + gv(a_0)$, coverage 63.214% against $1 - e^{-1} = 63.212\%$), and prove that $\text{rank}(\sigma_0 \oplus I) = 31$ gives a unique mask $\lambda = \text{0xa8a42fe4}$ with $\lambda \cdot (\sigma_0(W) \oplus W) = 0$ for every W (verified exhaustively over all 2^{32} inputs), which modular subtraction degrades to a bias of only $2^{-15.8}$.

Because these are properties of the parameterisations we tried rather than proofs of impossibility, we also build a reduced-width scale model in which the whole inner space is enumerable, and report exhaustive measurements there. **None of this is an impossibility result**, and we are explicit throughout about which claims are algebraic and which are statistical models that remain untested at full width. Full SHA-256 (64 rounds) is not threatened by any of these results.

Keywords: SHA-256, preimage attack, reduced-round, round-20 barrier, absorber tables, reduced-width scale model.

1 Scope and Relation to the $R = 19$ Paper

We assume the attack of [1] throughout: a backward chain from the target fixes $a_{12}, \dots, a_{19}$ (for $R=20$), the context words $a_4, \dots, a_{10}$ are freely chosen (one of the two admissible frames, §3.2), and the remaining unknowns are resolved against the schedule-consistency constraints C0–C3 at $W_{16}, \dots, W_{19}$ using a 16 GB global table inverting $u \mapsto \sigma_0(u) - u$. Notation follows that paper without change.

Nothing here modifies the $R=19$ result. The material is separated because it is exploratory: it consists largely of negative results, and its value is in narrowing where a $R=20$ attack could come from, not in delivering one.

2 The $R = 20$ Constraint System

The backward chain at $R = 20$ recovers $a_{12}, \dots, a_{19}$ but not a_{11} . The schedule grows to four constraints W_{16} – W_{19} , with a_{11} as an additional unknown.

Oracle-conditioned result. When the context including a_{11} is supplied by a known synthetic pair, the W_{17} identity gives a_2 as an explicit function of a_3 (Lemma analogous to [1, Prop. 2]), and an h -table $h[W_{12}(a_{11})] \rightarrow a_{11}$ enables an $O(2^{32})$ a_3 sweep. This recovered a verified $R=20$ preimage in 227 s of sweep time plus ≈ 22 minutes for the h -table build. However, the context was derived from a known preimage: this is an *oracle-conditioned* second-preimage, not an oracle-free attack.

Why oracle-free $R=20$ is not established. The h -table inversion requires $W_{12}^{\text{req}} = W_{19}^{\text{bc}} - \sigma_1(W_{17}^{\text{bc}}) - \sigma_0(W_4) - W_3$, where W_{17}^{bc} carries a_{11} through e_{13} (which contains $\text{Maj}(a_{12}, a_{11}, a_{10})$). Thus a_{11} is needed to compute the quantity used to look up a_{11} : a circular dependency. No algebraic link that breaks this circularity has been found. Exhaustive search over all binary linear combinations of the four residual equations R_{16} – R_{19} found no combination that isolates any of the three unknowns (a_2, a_3, a_{11}) .

Quantitative complexity gap. An equivalent view using the σ_0 -chain directly confirms the 2^{32} difficulty increase. Treat $a_4, \dots, a_{11}$ as freely chosen context (the backward chain fixes $a_{12}, \dots, a_{19}$). The chain C0–C2 then proceeds identically to the $R = 19$ case, but the schedule now includes W_{19} , introducing a fourth constraint (C3):

$$\sigma_0(W_4^{\text{round}}) = W_{19}^{\text{bc}} - \sigma_1(W_{17}^{\text{bc}}) - W_{12}^{\text{base}} - W_3,$$

where W_4^{round} is fully determined by $a_0, \dots, a_3$ once C0–C2 converge and W_{12}^{base} is determined by context. Because σ_0 is a bijection, W_4^{round} must equal the unique $\sigma_0^{-1}(C3_{\text{tgt}})$. C3 therefore contributes an additional 32-bit equality. Under the present construction we model it as approximately uniform and independent of C0–C2, so that it passes with probability $\approx 2^{-32}$; we stress that this is a statistical model which has not been established directly at full width.

Exhaustive experiments in a reduced-width scale model (§2.1) bear on the model but do not confirm it at $w = 32$. Measuring $P(\text{C3} \mid \text{C0}, \text{C1}, \text{C2})$ against the independent prediction 2^{-w} gives a ratio of 1.075 ± 0.018 at $w = 4$ and 1.005 ± 0.035 at $w = 5$: consistent with exact independence at the larger width, with a small excess at the smaller one that may be a narrow-width artefact. Any dependence is thus small, but two widths do not establish a trend, and the behaviour at $w = 32$ remains untested.

Schedule identity. For messages satisfying the σ_0 -chain $W_k = \sigma_0(W_{k+1})$, $k = 0, \dots, 3$, the substitution $\sigma_0(W_{r-15}) = W_{r-16}$ reduces the schedule recurrence for $r \in \{16, \dots, 19\}$ to

$$W_r = \sigma_1(W_{r-2}) + W_{r-7} + 2W_{r-16}.$$

From this identity and $W_3 = \sigma_0(W_4)$ one computes directly $C3_{\text{tgt}} = W_{19} - \sigma_1(W_{17}) - W_{12} - W_3 = \sigma_0(W_4) = W_3$, confirming that the C3 check reduces to $W_4^{\text{round}} = W_4^{\text{chain}}$ (verified

numerically). The same identity provides a concise derivation of the Nine-Step Lemma: at $r = 16$, $\Delta W_{16} = \Delta W_9 + \Delta W_0 = -\delta + \delta = 0$, and the doubled term $2W_{r-16}$ ensures that *any* perturbation $\Delta W_1 = \varepsilon$ would introduce $\sigma_0(\varepsilon)$ into ΔW_{16} (breaking cancellation), which is why the pair $(0, 9)$ is the unique minimal solution.

Orbit structure of σ_0 . Viewed as a $\mathbb{F}_2$ -linear bijection on $(\mathbb{Z}/2\mathbb{Z})^{32}$, the map σ_0 has order $49,146 = 2 \cdot 3 \cdot 8191$ where $8191 = 2^{13} - 1$ is a Mersenne prime. Its only short orbits are the two fixed points $\{0, 0x27f42515\}$ and the unique 2-cycle $\{0x0b3b7ef9 \leftrightarrow 0x2ccf5bec\}$; in particular, no period-3 or period-4 elements exist. Consequently any σ_0 -chain of length four $(W_0, \dots, W_4)$ lies in an orbit of length 49,146 unless W_4 is one of the four exceptional values above, so no “chain-wrap” shortcut of the form $W_0 = W_4$ arises generically.

Under the variable ordering and single-representative lookup construction used here, C3 remains an additional 32-bit consistency condition applied after C0–C2 have been solved. We emphasise what is and is not established. The algebra shows C3 imposes a further 32-bit equality; it does not show that every algebraic arrangement must pay 2^{32} for it. No absorption of C3 into the construction, and no lower-width decomposition of it, is currently known—but the negative results below are properties of the parameterisations we tried, not proofs of impossibility. Indeed one apparent obstruction at $R=20$, the $a_1 \leftrightarrow a_2$ cycle, dissolved entirely under a change of variable (§2.2), which is direct evidence that such obstructions can be artefacts of representation. Treating C3 as an independent filter, the cost is: Using the measured per-context success rate the projected $\mu_{R=19} \approx 6 \times 10^{-3}$ ([1, §6]), the $R=20$ expected contexts is $\mu_{R=19}^{-1} \times 2^{32} \approx 170 \times 2^{32} \approx 2^{39.4}$ —a factor of 2^{32} harder than $R=19$. The ratio is what matters here and is independent of the value of $\mu_{R=19}$; only the absolute figure moves with it, and that figure inherits the projection’s caveats.

GPU diagnostic. A 165-context GPU run of the $R = 20$ σ_0 -chain on an H100 (batch = 2^{26} , $K_{\text{seeds}} = 4$) recorded 4,166 W_9 -lo-pass events and zero hi-pass events. Under the null hypothesis of uniform residuals, the expected hi count is $\mu = 4166/65536 \approx 0.064$; the probability of zero hi events is $e^{-\mu} \approx 0.94$, so the absence of hi events is unsurprising. With an expectation below 0.1, however, observing zero carries almost no information: the result is *consistent with* a uniform 2^{-32} filter but does not test it. The observed difficulty increase over $R = 19$ is attributable to the additional C3 equality rather than to any structural bias in W_9 ; the attribution of the full 2^{32} factor to C3 rests on the uniformity model stated above, which this run is too small to test. A supplementary CPU sweep (10 contexts, batch = 2^{24} , two workers) recorded 254 W_9 -lo-pass events and zero C3-pass events, yielding an enhancement upper bound of $2^{26.3}$ (95%). This too is consistent with a uniform filter and far too small a sample to test independence or estimate its probability. The structural claim—that C3 imposes an additional 32-bit equality—rests on the algebra, not on these counts; the probabilistic claim should be read as untested.

2.1 A Reduced-Width Scale Model

Claims about C3’s independence, and about whether the chained solver is lossy, are untestable at full width: the events are too rare to sample and the solution set cannot be enumerated. We therefore build a scale model of the round function and message schedule at word width w , where the inner space $(a_0, a_1, a_2, a_3, a_{11})$ is small enough to enumerate exhaustively. Rotation and shift amounts are scaled from SHA-256’s by $w/32$ and rounded; K and the IV are truncated. The construction preserves the shape of the mixing and is not proposed as a standard.

Two fidelity checks are passed. At $w = 32$ the scaled constants reduce to SHA-256’s exactly, and the model reproduces SHA-256’s state evolution word for word. The solution count per context matches the degrees-of-freedom prediction: five unknown words less four constraints leaves one surplus word, hence 2^w solutions, against 19.2 measured at $w = 4$ ($2^4 = 16$) and 32.0 at $w = 5$ ($2^5 = 32$).

The independence measurement is reported in §2. The model also shows that the chained fixed-point architecture is intrinsically lossy rather than merely slow. A solver using one repre-

sentative per table entry recovers 1.3%, 0% and 0% of the true solutions at $w = 4, 5, 6$, while one branching over all roots recovers 23.4%, 6.8% and 0.7%. Raising the beam from 64 to 1024 and the iteration count from 6 to 20 changed coverage not at all, so the loss is structural rather than a truncation effect. Tables should therefore be relations, not functions. This monotone collapse, resting on 36, 13 and 2 recovered solutions, is the model’s robust finding.

We deliberately do *not* extrapolate these rates to $w = 32$. The per- a_0 success rate falls by $4.15\times$ from $w = 4$ to $w = 5$ and by $8.67\times$ from $w = 5$ to $w = 6$, so the point estimates suggest an accelerating rather than log-linear decay; but the $w = 6$ figure rests on two events, whose exact Poisson interval puts the second ratio anywhere in $[2.40\times, 71.6\times]$, a range that contains the first. The data therefore constrain neither the functional form nor the rate, while a 5% change in the fitted rate moves a 27-bit extrapolation by a factor of 4.5. We note additionally that an earlier version of this model, containing an initial-state indexing error, produced an apparent agreement with our full-width bound that did not survive correction—a cautionary instance of how little such an extrapolation constrains. The scale model’s value lies in its exhaustive in-range comparisons, not in extrapolated rates.

2.2 Absorber Reparameterisation and the $a_1 \leftrightarrow a_2$ Coupling

The four residual equations can each be written in the $\sigma_0(u) - u$ form used by the R=19 table. Substituting the round-function expression $W_{10} = \text{CONST}_{10} - d(a_3) - a_2$ into the W_{17} identity and then eliminating a_2 via $a_2 = W_2 - F_{12}(a_1)$ yields:

Lemma 1 (Reparameterisation of C1). *Write $A(a_1) := a_1 + gv + F_{12}(a_1)$, a function of a_1 alone. Then*

$$\sigma_0(W_2) - W_2 = h_1(a_{11}) - \text{CONST}_{10} + d(a_3) - A(a_1) =: u,$$

so the target u decomposes additively into independent contributions from a_{11} , a_3 and a_1 .

Verified exactly on 2000/2000 random triples (a_1, a_3, a_{11}) . Taking u as the free variable inverts the map: W_2 is one table lookup, $a_1 = A^{-1}(K_1 - u)$ a second, and $a_2 = W_2 - F_{12}(a_1)$. Hence for fixed (a_3, a_{11}) each u determines *both* a_1 and a_2 in $O(1)$. The apparent circular dependency between C0 (which needs a_2) and C1 (which needs a_1) is therefore an artefact of the parameterisation, not a genuine fixed-point problem; this explains why naive iteration of that pair never converges. (A is a near-bijection: 3 collisions in 2×10^5 samples, so A^{-1} requires a bucketed table.)

Two further reductions place the remaining constraints in table form. In C2 the unknown a_3 cancels from its own target, since $e_7 = a_3 + T1_7$ and $W_3 = a_3 + f_{23}$:

$$\sigma_0(W_3) - W_3 = (W_{18} - \sigma_1(W_{16}) - W_{11}) - W_2 - f_{23} + T1_7.$$

In C3, the quantities e_{16}, e_{17}, e_{18} do not depend on a_{11} (they are fixed by the backward chain), so $\sigma_1(e_{18})$ and $\text{Ch}(e_{18}, e_{17}, e_{16})$ are per-context constants and

$$W_{19} = \text{CONST}_{19} - (a_{11} + T1_{15})$$

is *linear* in a_{11} ; only W_{17} and W_{12} carry nonlinear a_{11} -dependence into the C3 key.

2.3 A Context Family that Removes a_2 from C0

The variable a_2 reaches the C0 target by exactly two routes: the term $\text{Ch}(e_8, e_7, e_6)$ with $e_6 = a_2 + t_{16}$, and $\text{Maj}(a_4, a_3, a_2)$ inside t_{15} . Both are maskable by context choice. Setting $e_8 = \text{0xFFFFFFFF}$ (reachable by choosing a_8 , since $e_8 = a_4 + a_8 - \Sigma_0(a_7) - \text{Maj}(a_7, a_6, a_5)$) collapses $\text{Ch}(e_8, e_7, e_6)$ to e_7 ; setting $a_4 = a_3$ collapses $\text{Maj}(a_4, a_3, a_2)$ to a_3 . Measured over 500 random a_2 per trial, the number of distinct C0 targets is:

context condition	distinct C0 targets
random context	400/400
$e_8 = 0\text{x}\text{FFFFFFFF}$ alone	400/400
$a_4 = a_3$ alone	217, 489
$a_4 = a_3$ and $e_8 = 0\text{x}\text{FFFFFFFF}$	1, 1, 1, 1

Neither condition suffices alone; together they eliminate a_2 from C0 entirely, turning it into a two-input absorber $(a_3, a_{11}) \mapsto a_1$. This is the first demonstration that the R=20 constraint *structure* is shapeable by context choice rather than fixed. It does not by itself reduce the constraint count.

2.4 The R=19 Iteration Architecture Does Not Transfer Directly

The R=19 attack does not schedule its constraints acyclically: it too has a cycle (C0 requires W_9 , which requires a_2, a_3 , which come from C1/C2, which require a_1 from C0). It breaks the cycle by freezing W_9 at a provisional value, iterating $C1 \leftrightarrow C2$ to a fixed point, and paying one 32-bit self-consistency check. Convergence of that iteration is what satisfies C1 and C2.

Promoting a_{11} from guessed context to an unknown absorbed by C3 gives R=20 the same one-word surplus (5 unknowns $a_0..a_3, a_{11}$ against 4 constraints, versus R=19’s 4 against 3), and we verified that the true R=20 solution is a fixed point of all four absorber maps. The architecture therefore transplants in form. It does not transplant in efficacy. Measured with one harness, cold start, 8 iterations, no seeding, on the same hardware:

iteration	convergences / trials	rate per a_0
R=19 control (two-map, $C1 \leftrightarrow C2$)	1,790 / 33,554,432	5.34×10^{-5}
R=20 (four-map, C0–C3)	0 / 1,996,488,704	$< 1.50 \times 10^{-9}$ (95%)

The R=19 control reproduces the documented rate band, so the harness is sound and the R=20 null is a measurement rather than an artefact. In iteration-steps per preimage the classic sweep costs 2^{64} while the iterated variant costs $2^{35}/p$, so the break-even convergence rate is $p = 2^{-29} \approx 1.86 \times 10^{-9}$; the measured bound lies below it. We note the margin is modest—the 95% bound is a factor 1.24 under break-even—so the conclusion is that the iterated variant does not beat the classic sweep, not that it loses by a wide factor; a longer run would tighten the bound proportionally. The structural estimate is far more pessimistic than the bound: R=19’s 5.34×10^{-5} sits about 2^{15} above the 2^{-29} random-map baseline for a 32-bit state, and carrying that same enhancement to a 64-bit state (baseline $\approx 2^{-61}$) predicts $p \approx 2^{-46}$, some five orders of magnitude below break-even. The cause is dimensional: at R=19 the absorbed a_1 is computed *outside* the loop, leaving a 32-bit iterated state (a_3 , with a_2 derived), whereas at R=20 C0 depends on a_{11} and the state becomes 64-bit (a_3, a_{11}). Raising the iteration budget does not help, since convergence probability and cost scale alike. Section 3.3 gives the structural reason the state cannot be held at 32 bits in *any* frame: at R=20 the full-avalanche dependency graph among the absorbed words acquires a cycle, and a cycle with no Maj/Ch-only edge has no cheap cut.

2.5 A Second Global Table: the a_0 Absorber

The attack sweeps a_0 and uses C0 to absorb a_1 through the global $\sigma_0(u) - u$ table. The two roles can be exchanged. Writing gv for the round-1 constant and using

$$W_1 = a_1 + gv(a_0), \quad W_0 = a_0 + C0_{\text{CONST}}, \quad W_9(a_1) = W_9^{(0)} - a_1,$$

where $W_9^{(0)}$ is W_9 evaluated at $a_1 = 0$, the constraint $W_{16} = \sigma_1(W_{14}) + W_9 + \sigma_0(W_1) + W_0$ rearranges to give:

Proposition 1. Let $\Gamma(a_0) := a_0 + gv(a_0)$ and $R := W_{16} - \sigma_1(W_{14}) - W_9^{(0)} - C0_{\text{CONST}}$. Then $C0$ is equivalent to

$$\Gamma(a_0) = R + W_1 - \sigma_0(W_1).$$

The point is that gv depends only on a_0 and the SHA-256 IV—no context word and no target enters it—so Γ is a fixed function and Γ^{-1} is a *global* table, built once and valid for every target and context, exactly like $\sigma_0(u) - u$. Measured over all 2^{32} inputs, Γ covers 63.214% of the range against $1 - e^{-1} = 63.212\%$ for a random map, so the table behaves like the one already in use. Proposition 1 is verified on all three published preimages by `code/verify_a0_absorber.py`, which also confirms it is not a tautology: perturbing W_{16} by $1, \dots, 64$ breaks it in every case.

This does not reduce the attack’s cost, and the reason is worth stating because it sharpens the $R = 20$ barrier. $C0$ can absorb a_0 or a_1 , but not both. With five unknowns ($a_0, a_1, a_2, a_3, a_{11}$) and four constraints, and with a_2 absorbable only by $C1$, a_3 only by $C2$ and a_{11} only by $C3$, the two variables a_0 and a_1 compete for the single constraint $C0$. One of them must therefore be swept, and whichever four remain form a cycle in which every absorber requires three of the others. That competition, rather than any property of a particular variable ordering, is what forces two guessed words at $R = 20$.

2.6 A Linear Identity for σ_0 , and its Destruction

Since σ_0 is $\mathbb{F}_2$ -linear, put $L = \sigma_0 \oplus I$. The kernel of L is the fixed-point set $\{0, 0x27f42515\}$, so $\text{rank } L = 31$ and the image is a 31-dimensional subspace. Consequently there is a *unique* nonzero mask annihilating it.

Proposition 2. Let $\lambda = 0xa8a42fe4$. Then for every $W \in (\mathbb{Z}/2\mathbb{Z})^{32}$,

$$\lambda \cdot (\sigma_0(W) \oplus W) = 0.$$

Verified exhaustively over all 2^{32} inputs (zero violations). This is the left companion of the known right-kernel constant $0x27f42515$.

The identity is however destroyed by modular subtraction, which is what the attack actually uses. Exhaustively over all 2^{32} inputs,

$$\Pr_W[\lambda \cdot (\sigma_0(W) - W) = 1] = 0.500017190,$$

a bias of $1.72 \times 10^{-5} = 2^{-15.8}$, against an exact 0 for the XOR form. A relation that is identically zero on the $\mathbb{F}_2$ -linear part survives modular subtraction only as a $2^{-15.8}$ bias, which affords no usable filtering. Together with the table coverage—both the σ_0 table and the $C3$ h -table measure 63.21% against $1 - e^{-1} = 63.212\%$, matching a random map to four significant figures—the map $W \mapsto \sigma_0(W) - W$ is indistinguishable from a random map under every structural test we have applied. The interaction between $\mathbb{F}_2$ -linear σ_0 and modular arithmetic is thus not merely an obstacle to proof technique; it is the mechanism by which $R=20$ resists.

The oracle-free $R=20$ attack remains an open problem.

3 The Absorber Assignment, and the Heavy-Cycle Barrier

This section gives the sharpest statement of the barrier we have, in two steps. First an inventory of which absorptions admit a *global* table, correcting an omission in the version of this inventory we circulated earlier. Second a measurement of the dependency structure among the absorbed words that explains, with one number per edge, why the $R=19$ iteration architecture cannot be carried to $R=20$ in any frame. Unlike the earlier “irreducible cycle” claim—which was wrong, and which we withdraw—neither step is an argument about cycles alone: the $R=19$ assignment is also cyclic and succeeds. What matters is the *weight* of the edges.

3.1 Linear Entries and the Global Absorbers

Write $T_2(x, y, z) = \Sigma_0(x) + \text{Maj}(x, y, z)$. In the a -only representation of [1], $e_r = a_{r-4} + a_r - T_2(a_{r-1}, a_{r-2}, a_{r-3})$ and the recovered message word is

$$W_r = a_r - T_2(a_{r-1}, a_{r-2}, a_{r-3}) - e_{r-4} - \Sigma_1(e_{r-1}) - \text{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) - K_r.$$

Lemma 2 (Linear entries). *For every state word a_k : $W_k = a_k + f(a_{k-8}, \dots, a_{k-1})$; $W_{k+8} = -a_k + g(a_{k+1}, \dots, a_{k+8})$; a_k enters $W_{k+1}, \dots, W_{k+7}$ nonlinearly; and it enters no other W_r .*

Proof. W_r depends on $a_{r-8}, \dots, a_r$ and nothing else (the nine-word dependence of [1]). In W_k the word a_k occurs once, as the leading term. In W_{k+8} it occurs once, inside the h -term $e_{k+4} = a_k + a_{k+4} - T_2(a_{k+3}, a_{k+2}, a_{k+1})$, which is subtracted. For $1 \leq m \leq 7$ it occurs only inside a T_2 , a Σ_1 or a Ch argument. $\square$

The lemma is checked numerically by `edge.weights.py`: for $k \leq 4$ at $R=20$ and three distinct increments each, W_k moves by exactly $+\delta$, W_{k+8} by exactly $-\delta$, the words between by neither, and $W_k + W_{k+8}$ is invariant under a_k .

Corollary 1 (Every constraint has a global absorber). $C_j = W_{16+j} - \sigma_1(W_{14+j}) - W_{9+j} - \sigma_0(W_{1+j}) - W_j$ contains a_{j+1} exactly twice: as $+a_{j+1}$ inside the σ_0 argument W_{1+j} and as $-a_{j+1}$ inside W_{9+j} . With $u = W_{1+j}$ it reads $\sigma_0(u) - u = (\text{terms free of } a_{j+1})$, so the single global table absorbs a_{j+1} from C_j for every $j = 0, \dots, R-17$.

For $j = 0, 1, 2$ this is the opposite-sign pattern that drives the $R=19$ attack. For $j = 3$ it is $C3 \rightarrow a_4$, which the inventory we circulated earlier listed as absent; the corrected inventory is:

Global (built once, valid for every target and context)	absorbs
$\sigma_0(u) - u$	$C_j \rightarrow a_{j+1}$ for every j : $C0 \rightarrow a_1, C1 \rightarrow a_2, C2 \rightarrow a_3, C3 \rightarrow a_4$
Γ^{-1} , where $\Gamma(a_0) = a_0 + gv(a_0)$ (§2.5)	$C0 \rightarrow a_0$
$W_0 = a_0 + \text{const}$ (linear)	—
Per-context (2^{32} to build, reusable within one context)	
$Q_0^{-1}, Q_1^{-1}, Q_2^{-1}, H^{-1}$	$C0-C3 \rightarrow a_{11}$
No table (context-dependent inversion, 2^{32} every time)	
$C3$ absorbing a_1, a_2 or a_3 ; $C0$ absorbing a_4	—

3.2 Two Frames, Both Cyclic

Reaching 2^{32} per context requires sweeping one unknown and absorbing four. Given the inventory there are exactly two ways to do it at $R=20$. *Frame A* (used in §2.4) keeps $a_4, \dots, a_{10}$ as context, sweeps a_0 , and absorbs a_1, a_2, a_3 globally and a_{11} per-context:

$$a_1 \leftarrow C0(a_2, a_3, a_{11}), \quad a_2 \leftarrow C1(a_1, a_3, a_{11}), \quad a_3 \leftarrow C2(a_1, a_2, a_{11}), \quad a_{11} \leftarrow C3(a_1, a_2, a_3).$$

Frame B keeps $a_5, \dots, a_{11}$ as context, sweeps a_0 , and absorbs $a_1, \dots, a_4$ with the one global table:

$$a_1 \leftarrow C0(a_2, a_3, a_4), \quad a_2 \leftarrow C1(a_1, a_3, a_4), \quad a_3 \leftarrow C2(a_1, a_2, a_4), \quad a_4 \leftarrow C3(a_1, a_2, a_3).$$

Sweeping something other than a_0 does not help in either frame, because a_0 enters $C0$ both inside $\sigma_0(a_1 + gv(a_0))$ and additively through W_0 , and separating those requires a table indexed by a_1 . Both assignments are cyclic. So is the $R=19$ assignment ($a_1 \leftarrow C0(a_2, a_3)$, $a_2 \leftarrow C1(a_1, a_3)$, $a_3 \leftarrow C2(a_1, a_2)$), and it succeeds; cyclicity is not the barrier.

3.3 Edge Weights: the Heavy Cycle Appears at $R = 20$

What distinguishes the three assignments is how strongly each absorbed word feeds back into the constraints that absorb the others. We measure it directly: for random targets, contexts and unknowns, flip one random bit of a_k and record the Hamming weight of the change in C_j , averaged over 200 trials per cell (`edge_weights.py`, seed 20260902; raw output in `data_edge_weights.txt`). Full avalanche is about 16; a *mild* edge, one that passes only through Maj and Ch of a_k plus a constant, moves one or two bits.

flip	$R=19$			$R=20$, frame B				$R=21$				
	C0	C1	C2	C0	C1	C2	C3	C0	C1	C2	C3	C4
a_0 (swept)	15.4	15.9	15.9	15.6	15.8	15.8	15.0	15.1	16.1	15.7	15.2	15.5
a_1	<u>8.0</u>	15.3	16.2	<u>7.6</u>	15.2	16.1	15.7	<u>8.0</u>	15.1	16.2	15.9	15.3
a_2	1.5	<u>7.9</u>	15.1	1.3	<u>7.9</u>	14.9	16.3	1.5	<u>7.6</u>	14.9	16.1	16.0
a_3	1.9	1.4	<u>8.3</u>	1.9	1.6	<u>8.3</u>	15.2	1.7	1.4	<u>7.8</u>	15.4	16.1
a_4				12.2	1.7	1.4	<u>7.8</u>	12.4	1.8	1.5	<u>7.8</u>	15.2
a_5								8.3	12.4	1.6	1.3	<u>8.0</u>

Underlined cells are self-edges (the absorbed word moving its own $\sigma_0(u) - u$ term); entries above the diagonal are forward edges; entries below it are feedback edges, with the heavy ones in bold. Frame A at $R=20$ (rows a_1, a_2, a_3, a_{11} , same script) has feedback weights $a_{11} \rightarrow C0$ 15.7, $a_{11} \rightarrow C1$ 13.3, $a_{11} \rightarrow C2$ 15.0.

Reading. At $R=19$ every feedback edge is mild. The full-avalanche (“heavy”) dependency graph is the chain $a_1 \rightarrow a_2 \rightarrow a_3$, which is acyclic, and the attack exploits exactly this: it computes a_1 outside the loop, iterates $a_3 \rightarrow a_2 \rightarrow a'_3$, and thereby cuts its cycle at the mild edge $a_3 \rightarrow C1$. That is why the iterated map is nothing like a random map (its convergence rate of 5.34×10^{-5} sits 2^{15} above the random-map baseline, §2.4). At $R=20$ exactly one feedback edge is heavy, $a_4 \rightarrow C0$ at 12.2 bits, and the heavy graph becomes the cycle $a_1 \rightarrow a_2 \rightarrow a_3 \rightarrow a_4 \rightarrow a_1$. A heavy cycle has no mild cut. Cutting it at any single 32-bit word composes full-avalanche maps with table lookups, which is a random map on 2^{32} points; its fixed points cost 2^{32} each to find, which is the classic sweep. Frame A is worse, with three heavy feedback edges through a_{11} . This is the content of the “dimensional” explanation in §2.4: the state cannot be held at 32 bits because no mild cut exists. It is also why partial seeding of (a_2, a_3, a_4) never helped in our early $R=20$ experiments: the seed had to be exact in a_4 , because a_4 is the word on the heavy edge.

Why it is the round function and not the frame. The weight of an entry is set by its offset. a_k enters W_{k+m} for $m = 6, 7$ only through Maj and Ch of a_k plus constants (mild); for $m = 5$ through the h -term e_{k+1} , which contains $\Sigma_0(a_k)$ subtracted directly, and through $\Sigma_1(e_{k+4}) = \Sigma_1(a_k + c)$ (heavy; measured 12.2–12.4); for $m = 4$ through Ch($e_{k+3}, e_{k+2}, e_{k+1}$), which selects about half the bits of $\Sigma_0(a_k)$ (measured 8.3); and for $m \leq 3$ through Σ_0 and Σ_1 of a_k itself. The last word of the window, a_n with $n = R - 16$, feeds C0 through W_9 at offset $9 - n$, which is mild if and only if $n \leq 3$, i.e. $R \leq 19$. No context choice removes $\Sigma_0(a_4)$ from W_9 , because no context word stands between them. At $R=21$ the window has three heavy feedback edges (table), and the count grows from there.

What would cut it. A trick that cancels the barrier must cut one heavy edge, and the cheapest candidate is the new one: in C0, invert $\Sigma_0(a_4) - \Sigma_1(a_4 + c_8)$ plus its Maj/Ch terms *jointly* with the a_1 absorber $\sigma_0(W_1) - W_1$ at well below 2^{32} per instance. All of these are rotation–XOR maps, so over GF(2) they would combine linearly; the modular carries between them are what destroy the combination, and §2.6 measures that destruction for σ_0 alone (an exact GF(2)

identity degraded to a $2^{-15.8}$ bias). The routes listed in earlier versions of this section—a normal form for $\sigma_0(W_4) + W_3$ in one message word, freeing the sweep variable through the a_0 absorber, a joint two-variable global table—are all instances of cutting a heavy edge, and none has been found. The two attempts recorded below are of this kind and both charge full price.

3.4 Attempts to Cut the Cycle

The one structural handle, and why it does not pay. C3 has the form $\sigma_0(W_4) + W_3 = H(a_{11})$, coupling *two* message words—this is precisely why it cannot absorb a_1, a_2, a_3 through the single-word table. Imposing any global relation $W_3 = \phi(W_4)$ collapses the left side to a function of one word, making C3 a global absorber producing (W_3, W_4) from a_{11} alone and reversing the cycle direction. Measured exhaustively in the reduced-width model at $w = 4$ over 6 contexts (98 true solutions; expectation 6.12 if ϕ costs a full w bits):

Relation	surviving	per context	vs. expected
$W_3 = \sigma_0(W_4)$	3	0.50	$0.49\times$
$W_3 = -W_4$	4	0.67	$0.65\times$
$W_3 = W_4$	0	0.00	$0.00\times$
$W_3 = \overline{W_4}$	7	1.17	$1.14\times$

Every variant sits at or below the “costs a full w bits” expectation. The counts are small (3–7 events, so roughly ± 2) and all are consistent with $1.0\times$; none is consistent with the large excess needed to make the handle pay. Imposing a w -bit condition removes a w -bit worth of solutions, and the global absorber it buys does not compensate.

Elimination orders. An elimination order cannot change an edge weight, so the heavy-cycle argument predicts that no order recovers the solution set. The exhaustive search over all 2,880 elimination orders in the reduced-width model ($w=4$, 41 true solutions across its contexts), repeated after the correction to the model’s initial-state indexing (§2.1), agrees: 2,560 of 2,880 orders find at least one solution, the median finds 2 and the best finds 11, and the hand-chosen $R=19$ -style order (sweep a_0 ; $a_1 \leftarrow C0$, $a_2 \leftarrow C1$, $a_3 \leftarrow C2$, $a_{11} \leftarrow C3$) ranks 175th with 5. The pre-correction run had reported a median of 2 of 35; the conclusion is unchanged.

4 Status

$R = 20$ is not solved. The barrier is 2^{64} for this attack family, and its sharpest form is the heavy-cycle argument of §3.3: the full-avalanche dependency graph among the absorbed words is acyclic for $R \leq 19$ and cyclic from $R=20$ in every admissible frame, because $\Sigma_0(a_k)$ is subtracted directly into W_{k+5} . Context screening ([1, §5.7]) raises W_9 lo-pass yield by a measured $3.41\times$ (95% CI [2.13, 5.82]) with conversion to preimages unmeasured; taken at face value that is about $2^{62.2}$, an improvement and not a solution.

We emphasise the limits of the negative results. Each rules out a parameterisation, not a technique: the ϕ -family charges a full word for the global absorber it buys; a_0 is globally absorbable but competes with a_1 for C0; a joint two-variable table is blocked because a_2 and a_3 enter shift-invariantly; no elimination order helps. The heavy-cycle argument is a measurement of this attack family, not a lower bound. What would change the picture is a way to invert $\Sigma_0(a_4) - \Sigma_1(a_4 + c_8)$ jointly with $\sigma_0(W_1) - W_1$ inside C0 at well below 2^{32} , and nothing measured here excludes one; it does say that any such trick is a statement about carries between rotation–XOR maps, which is where §2.6 found only a $2^{-15.8}$ bias.

References

- [1] K. S. Basra. A Practical Oracle-Free Preimage Attack on 19-Round SHA-256, and a Nine-Step Cancellation Lemma. Companion paper, 2026.